Quiz 1: Cloud Services and Deployment

1. B
2. C
3. C
4. C
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20. C

Quiz 2: Microservices and Docker

1. C
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6. C

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13. C
14. A
15. B
16. C
17. C
18. B
19. C
20. B

Activity 1: Cloud Service Model Comparison

Comparison Table of IaaS, PaaS, and SaaS

Feature	**IaaS (Infrastructure as a Service)**	**PaaS (Platform as a Service)**	**SaaS (Software as a Service)**
Description	Provides virtualized computing resources such as servers, storage, and networking over the internet	Provides a platform for developers to build, test, and deploy applications without managing infrastructure	Provides ready-to-use software applications accessible through the internet
User Manages	Applications, data, runtime, middleware, operating system	Applications and data	Only user settings and data
Provider Manages	Hardware, networking, virtualization	Infrastructure, operating system, runtime	Everything including application maintenance
Examples	Amazon EC2, Microsoft Azure VM, Google Compute Engine	Google App Engine, Heroku, AWS Elastic Beanstalk	Google Workspace, Microsoft 365, Dropbox

Advantages	Flexible, scalable, full control of infrastructure	Faster development, less infrastructure management	Easy to use, accessible anywhere, low maintenance
Limitations	Requires technical expertise and maintenance	Limited control over infrastructure	Limited customization and control
Best Use Case	Hosting websites and enterprise systems	Application development and deployment	Online productivity and collaboration tools

Real-World Scenarios and Justification

IaaS Scenario

- A startup company wants to host its e-commerce website but does not want to buy physical servers.

Justification: IaaS is suitable because it provides scalable virtual machines and storage resources. The startup can increase or decrease resources depending on customer traffic while only paying for what it uses.

PaaS Scenario

-A software development team is building a mobile application with multiple developers working together.

Justification: PaaS is ideal because developers can focus on coding and deployment without worrying about managing servers, operating systems, or infrastructure.

SaaS Scenario

-A school uses Google Workspace for online classes, email, and document sharing.

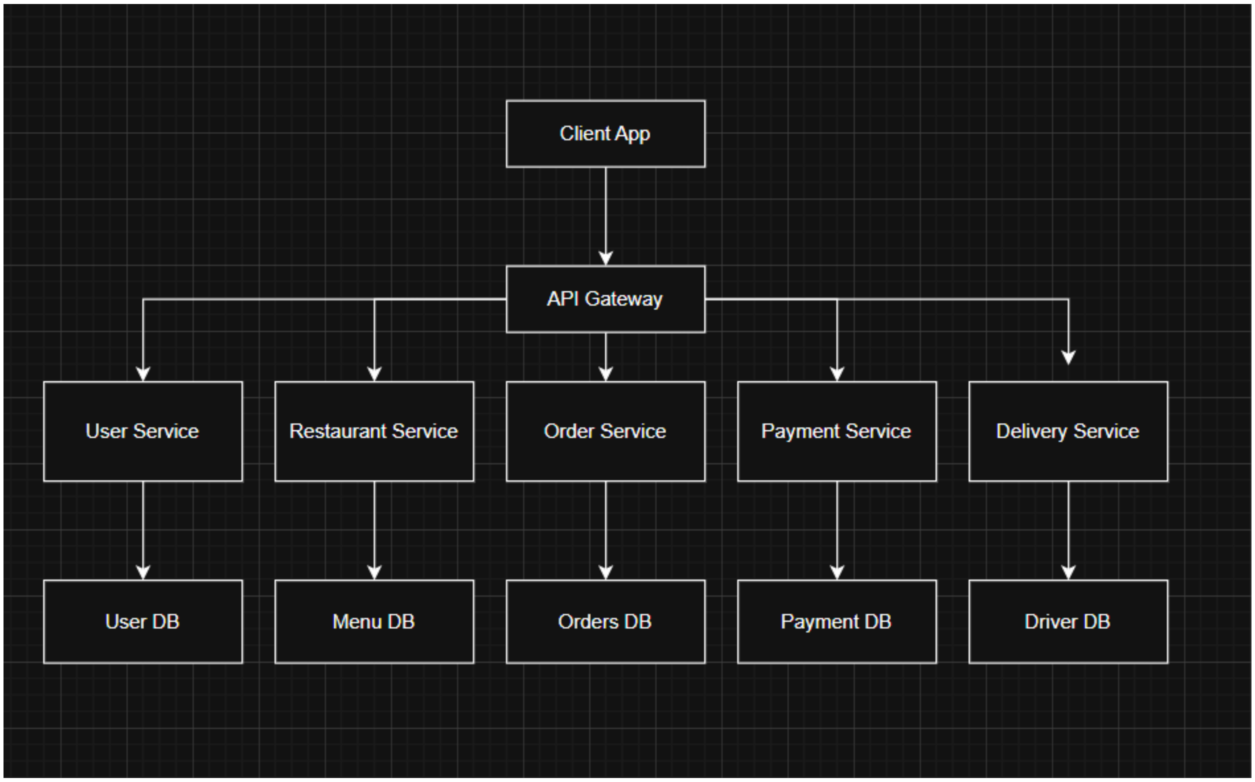
Justification: SaaS is appropriate because users can immediately access applications through a browser without installation or maintenance.

Activity 2: Microservices Architecture Design

Chosen Application: Online Food Delivery System

1. User Service
2. Restaurant Service
3. Order Service
4. Payment Service
5. Delivery Service

Architecture Diagram



Communication:
- Client to API Gateway: HTTP/REST
- Service-to-Service: REST API and RabbitMQ

Explanation of the Architecture
The Online Food Delivery System uses a microservices architecture where each service handles one business function.

API Gateway
The API Gateway acts as the main entry point for all client requests. It routes requests to the correct service and improves security and traffic management.

User Service
Handles user registration, login, and profile management.

Restaurant Service
Manages restaurant information, menus, and food availability.

Order Service
Processes customer orders and order status.

Payment Service

Handles online payments and transaction verification.

Delivery Service
Tracks delivery drivers and order delivery progress.

Communication Method
The system uses REST APIs for synchronous communication and RabbitMQ for asynchronous communication between services.

Failure Handling
If one service fails, the other services can continue operating independently.
Example: If the Payment Service becomes unavailable:

- Customers can still browse restaurants and place orders.
- The Order Service can temporarily mark orders as “Pending Payment.”
- RabbitMQ queues can store payment requests until the service recovers.
- Monitoring tools can alert administrators about the failure.